Failure Analysis of Adhesives

Kashyap, Anuj
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***Abstract*—Using measured failure load and post-failure adhesive area, this experiment assessed the shear strength of four adhesives in 6061-T6 aluminum double-lap joints. Each lab section produced and tested a single specimen of each adhesive, and the failed bond area for each specimen was calculated using calibrated ImageJ pictures following fracture. The measured maximum load and the total of the two failed adhesive areas were used to determine the maximum shear stress. Specimens that did not sufficiently satisfy the equal-load assumption for statistical comparison were identified using a relative-error check. AA H3000 achieved an average maximum shear stress of 20 MPa with a 95% confidence interval of 20±4.2 MPa, making it the strongest adhesive tested. Average strengths of 6.4, 4.5, and 3.3 MPa were significantly lower for EA E60HP, CA, and DP640, respectively. This pattern was corroborated by failure observations, which showed that cohesive behavior was present in the stronger AA H3000 group whereas adhesive failure predominated in the lower-strength groups. When compared to reference data, AA H3000 nearly matched its predicted shear-strength value, but DP640 and EA E60HP fared worse than their stated reference values. Overall, the findings demonstrated that the choice of adhesive had a significant impact on joint strength; however, the assessed performance was also significantly influenced by specimen preparation, bonded-area symmetry, and failure mode.**

***Index Terms*—Adhesive joints, double-lap shear, ImageJ, shear strength**

I. Introduction

The objective of this experiment was to determine the shear strength of four common adhesive families in double-lap shear joints made from 6061-T6 aluminum [1], [2]. The adhesives evaluated were DP640, AA H3000, EA E60HP, and CA adhesive [1]. One specimen of each adhesive was fabricated and tested in each lab section so that the measured joint performance could be compared across sections and later compared with expected manufacturer behavior [2]. In this lab, the reported shear strength of each specimen was determined from two measured quantities: the maximum load carried by the specimen during testing and the adhesive bond area that failed during fracture [1], [2].

Given that adhesives can join similar or dissimilar materials without creating the local stress concentrations associated with bolts or rivets, and because they work best when the applied load is transferred primarily through shear rather than direct tension, adhesive joints are crucial in engineering [1]. Differentiating between cohesive failure, which happens inside the glue, and adhesive failure, which happens at the interface between the adhesive and the parent material, is also crucial in adhesive-joint analysis [1]. This distinction was crucial in the current experiment because the look of the failed surfaces helped determine whether the adhesive substance or the adhesive-aluminum bond was the primary factor limiting the measured joint strength. Even when the same parent material was used significant specimen-to-specimen variation was anticipated since adhesive effectiveness also depended on surface cleanliness, surface roughness, ambient conditions, and cure history [1].

To load the adhesive primarily in shear, the joints were tested using a four-strip double-lap design [1], [2]. To concentrate failure in a specified area, one bonded overlap in this shape was purposefully made smaller than the other [1]. This reduced-overlap design decreased uncertainty in the post-failure area measurement and facilitated the identification of the desired failure point following fracture. The four-strip configuration was chosen because it is simpler to grip and loads more symmetrically in the testing machine, even though a three-strip geometry might more precisely specify the adhesive area [1]. The specimen, as shown in Fig. 1, was made up of four overlapping strips with the labels A, B, C, and D. The greater overlap b_2 preserved the overall double-lap arrangement, whereas the lower overlap length b_1 was employed to concentrate failure in the reduced bonded region.

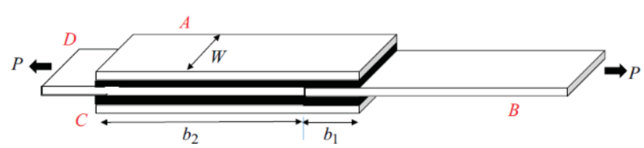

Fig 1. Geometry of the double-lap shear specimen used in Lab 4. The specimen consisted of four overlapping 6061-T6 aluminum strips, where the reduced overlap b_1 was intended to promote failure in a controlled bonded region and the larger overlap b_2 maintained the double-lap load path.

The two load pathways, D→A→B and D→C→B, were considered to carry equivalent loads in the analysis [1]. According to this assumption, the failure load P, and the total failed shear area 2A determine the nominal shear stress at failure, τ_{max}, which is expressed in MPa, P in N, and area in mm^2. This connection is provided by equation 1.

$$\tau_{max} = \frac{P_{max}}{2A} = \frac{P_{max}}{A_A + A_C} \tag{1}$$

The total failed shear area was calculated from the sum of the two measured failure areas since the two failed outer adhesive surfaces were measured independently following testing. Let A_A and A_C represent the failed adhesive areas on strips A and C, respectively, in millimeters. Thus, the overall failed shear area was determined to be.

$$A_{shear} = A_A + A_C \tag{2}$$

Using the measured maximum load $P_{\max}$ and the combined failed area from both outer strips, the working equation used in the report becomes.

$$\tau_{max} = \frac{P_{max}}{A_{shear}} \tag{3}$$

Since this model depends on approximate symmetry between the two bonded sides, the validity of the assumption was also checked using a worst-case estimate based on the smaller of the two measured failed areas. The shear stress that results from using the smaller bonded area as the controlling interface is represented by τ_{worst}. The sensitivity of the outcome to area imbalance was assessed using the relative difference between this worst-case value and the nominal value:

$$Relative\ Error = \frac{\tau_{worst} - \tau_{max}}{\tau_{max}} \tag{4}$$

$$\tau_{worst} = \frac{P}{2 \cdot \min{(A_A, A_C)}} \tag{5}$$

The assumed symmetric load transfer through the joint is illustrated in Fig. 2. This figure supports the equal-load assumption used in Eqs. (1) through (5) and shows why each bonded side was treated as carrying one-half of the applied load [1].

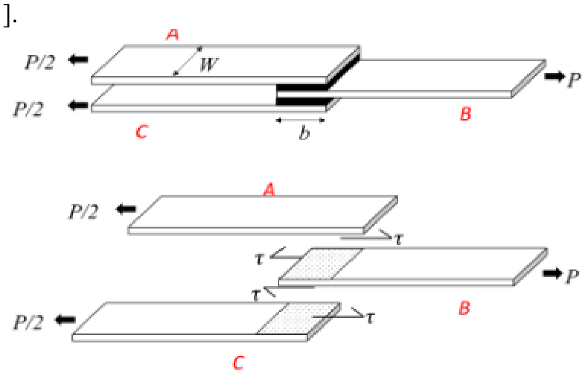

Fig 2. Idealized load paths and shear-force transfer in the double-lap specimen. Under the equal-load assumption, the applied load P is divided equally between the two bonded interfaces, so each shear plane carries P/2 and develops an average shear stress τover the bonded region of length b.

The maximum load that each joint could support was recorded after the specimens were tested to failure in the universal testing system at a crosshead displacement rate of 7 mm/min [2]. Because adhesive joints might break suddenly, displacement control was necessary. By prescribing crosshead motion, the load could increase steadily while still recording the peak failure load. Following fracture, each specimen was disassembled, two consensus failure areas were measured in ImageJ, and the failed surfaces were photographed using a scale for calibration [2]. The nominal shear stress at failure was then calculated using Eq. (3) and the sum of these two measured regions. Because one specimen of each adhesive was tested in each lab section, the resulting dataset also supported comparison across sections through 95% confidence intervals and comparison of statistical variation with propagated force and area uncertainty [1], [2].

II. Procedure

A. Specimen Fabrication

The Four double-lap shear specimens made of 6061-T6 aluminum were put together during the first week of Lab 4. One specimen was made for each type of glue, including cyanoacrylate, polyurethane, methacrylate, and epoxy [1], [2]. To enhance adhesion by eliminating surface impurities and preparing the bonding surfaces for the adhesive, the aluminum surfaces were lightly sanded and washed with alcohol before bonding [1]. After applying the adhesive, the strips were put together to fit the four-strip double-lap shape shown in Figure 1. During testing, the reduced overlap was utilized to concentrate failure in a known bonded region [1], [2]. The strips were aligned symmetrically during assembly and held in position during initial hardening so that the intended joint geometry was maintained [2]. After assembly, the specimens were left undisturbed to cure until the second week of Lab 4. Each specimen was also marked with match marks and a section identifier so that the strips could be identified and reassembled correctly after fracture [2].

B. Mechanical Testing

Before The four section specimens underwent individual testing in the universal testing system during the second week of Lab 4 [2]. Displacement control was used to load each specimen to failure at a crosshead rate of 7 mm/min after the double-lap joint was aligned in the grips [2]. Each specimen's maximum load was noted and saved for use in subsequent shear-stress computations [1], [2]. This loading strategy was chosen since the lab's goal was to compare the four adhesive systems' shear performance using the failure load that the testing apparatus directly measured [2].

C. ImageJ Area Measurements

Each specimen was separated upon failure to reveal the adhesive surfaces that had failed [2]. The individual strips and the associated failure locations were identified using the match markings that were applied during manufacture [2]. For the image to be spatially calibrated in ImageJ, each failed surface of interest was subsequently shot with a scale added in the image [2]. By properly photographing the failure surfaces and integrating the calibration scale with the specimen, care was taken to reduce uncertainty in the area measurement [2]. Each specimen was assumed to have two failure surfaces, and two consensus area estimates were given [2]. The adhesive bond area that failed in shear during testing was represented by these measured areas [1], [2].

D. Data Analysis

The two consensus failure areas measured in ImageJ and the highest load recorded during UTS testing for each specimen were the retained outputs used in the subsequent study [1], [2]. These values were represented by the symbols Pmax, AA, and AC, where Pmax is the maximum failure in N and A_A and A_C are the measured failed adhesive areas on the two outside strips in mm2. The shear-strength calculation used the sum A_A+A_C as the overall failed shear area. The resulting results were then utilized for the necessary statistical and uncertainty analysis as well as comparing the adhesive performance across sections [1], [2].

III. Results

The two failed adhesive areas (AA and AC) and the maximum load at failure (Pmax) for each specimen were the directly measured parameters in this experiment. The raw measured results are shown below by adhesive type because only one specimen of each adhesive was evaluated during each lab time. The computed shear strengths and subsequent statistical screening are saved for the Discussion section of the current report, while the Results section is limited to the measured regions, measured loads, and representative failure photographs.

A. DP640 Adhesive

Table I lists the DP640 adhesive's measured failure areas and maximum loads. The measured side C areas were between 352 and 819 mm2, while the measured side A areas were between 307 and 867 mm^2. The greatest loads that were measured varied between 939 and 5498 N. On both measured sides, the majority of DP640 specimens were between 350 and 590 mm^2, with lab period R 9_35 having the biggest measured areas. The DP640 specimens were primarily categorized as adhesive failures based on the documented failure labels.

TABLE I
Area and Load Measurements for DP640 Adhesive

Lab Period	A_A (mm^2)	A_C (mm^2)	P_{Max} (N)
T 9_35	354 ± 7.5	352 ± 7.5	3365 ± 17
T 11_45	378 ± 8.0	389 ± 8.2	3581 ± 18
T 1_55	480 ± 10	433 ± 9.2	5498 ± 27
T 4_05	355 ± 7.5	356 ± 7.5	1645 ± 8.2
W 10_40	307 ± 6.5	352 ± 7.5	1299 ± 6.6
W 12_50	459 ± 9.7	526 ± 11	3405 ± 17
W 3_00	497 ± 11	495 ± 10	939 ± 4.7
R 9_35	867 ± 18	819 ± 17	5064 ± 25
R 11_45	446 ± 9.4	449 ± 9.5	2593 ± 13
R 1_55	456 ± 9.7	454 ± 9.6	3295 ± 16
R 4_05	587 ± 12	562 ± 12	3013 ± 15

A representative DP640 failure surface is shown on Fig. 3. This figure should show the photographed fracture surface used for the post-failure ImageJ area measurement.

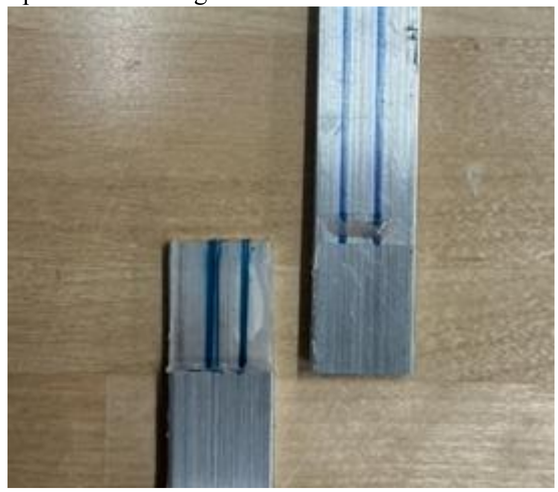

Fig 3. Failure surface of DP640 from R1_55 Lab.

B. AA H3000 Adhesive

Table II lists the AA H3000 adhesive's measured failure areas and maximum loads. The side C areas were measured between 243 and 779 mm^2, and the side A areas were measured between 202.563 and 594.400 mm^2. Over the course of the full set of lab periods, the recorded maximum loads, which varied from 7485 to 18576 N, were greater than those of the other adhesive groups. Additionally, the recorded failure mode varied the most among the AA H3000 specimens. Several specimens showed cohesive failure, lab periods T 11_45, W 12_50, and R 1_55 showed mixed failure, and lab period R 4_05 showed substrate failure.

TABLE II
Area and Load Measurements for AA H3000 Adhesive

Lab Period	A_A (mm^2)	A_C (mm^2)	P_{Max} (N)
T 9_35	433 ± 9.2	337 ± 7.1	14539 ± 73
T 11_45	425 ± 9.0	399 ± 8.5	15386 ± 77
T 1_55	594 ± 13	779 ± 17	14844 ± 74
T 4_05	274 ± 5.8	271 ± 5.7	13999 ± 70
W 10_40	425 ± 9.0	422 ± 8.9	17849 ± 89
W 12_50^x	426 ± 9.0	244 ± 5.2	7485 ± 37
W 3_00^x	203 ± 4.3	300 ± 6.4	16424 ± 82
R 9_35	408 ± 8.6	435 ± 9.2	17995 ± 90
R 11_45	446 ± 9.4	447 ± 9.5	18398 ± 92
R 1_55^x	469 ± 9.9	243 ± 5.1	16726 ± 84
R 4_05^y	523 ± 11	521 ± 11	18576 ± 93

^{x} Removed in later calculations due to large relative error.
^{y} Removed in later calculations due to substrate failure.

A representative AA H3000 failure surface is shown in Fig. 4. This figure should document the fracture surface photographed after specimen disassembly and be used for the corresponding area estimate.

Fig 4. Failure surface of AA H3000 from R1_55 Lab.

C. EA E60HP Adhesive

Table III lists the EA E60HP adhesive's measured failure areas and maximum loads. The measured side C areas varied from 152 to 536 mm^2, while the measured side A areas varied from 151 to 537 mm^2. The maximum loads that were measured varied between 1227 and 12887 N. The EA E60HP findings revealed a wide range in both measured area and measured load, with lab period R 9_35 exhibiting the highest recorded load. While mixed failure was noted in lab periods R 9_35 and R 11_45, the majority of EA E60HP specimens were categorized as adhesive failures.

TABLE III
AREA AND LOAD MEASUREMENTS FOR EA E60HP ADHESIVE

Lab Period	A_A (mm^2)	A_C (mm^2)	P_{Max} (N)
T 9_35	480 ± 10	415 ± 8.8	3967 ± 20
T 11_45	151 ± 3.2	152 ± 3.2	3068 ± 15
T 1_55	225 ± 4.8	254 ± 5.4	1227 ± 6.1
T 4_05	438 ± 9.3	342 ± 7.2	5087 ± 25
W 10_40	480 ± 10	428 ± 9.1	7781 ± 39
W 12_50	537 ± 11	483 ± 10	6416 ± 32
W 3_00	424 ± 9.0	450 ± 9.5	7464 ± 37
R 9_35	515 ± 11	536 ± 11	12887 ± 64
R 11_45	442 ± 9.4	436 ± 9.2	6576 ± 33
R 1_55	460 ± 9.8	478 ± 10	5927 ± 30
R 4_05	485 ± 10	440 ± 9.3	3363 ± 17

A representative EA E60HP failure surface is shown on Fig. 5. This figure should present the photographed fracture surface used in the post-failure area measurement.

Fig 5. Failure surface of EA E60HP from R1_55 Lab.

D. CA Adhesive

Table IV lists the CA adhesive's measured failure areas and maximum loads. The measured side C areas were between 202 and 635 mm2, and the measured side A areas were between 277 and 814 mm2. The greatest loads that were recorded varied from 1710 to 16726 N. Among the four glue groups, the CA data displayed one of the largest spreads in recorded maximum load. The R 1_55 specimen recorded a significantly higher maximum load than the other CA specimens, while most epoxy specimens failed to reach roughly 4300 N. The CA specimens were primarily categorized as adhesive failures based on the documented failure labels.

TABLE IV
AREA AND LOAD MEASUREMENTS FOR CA ADHESIVE

Lab Period	A_A (mm^2)	A_C (mm^2)	P_{Max} (N)
T 9_35	417 ± 8.8	410 ± 8.7	1744 ± 8.7
T 11_45^x	814 ± 17	481 ± 10	4302 ± 22
T 1_55	277 ± 5.9	202 ± 4.3	1710 ± 8.5
T 4_05	465 ± 9.9	463 ± 9.8	2808 ± 14
W 10_40	488 ± 10	499 ± 11	2344 ± 12
W 12_50	503 ± 11	390 ± 8.3	2483 ± 12
W 3_00	573 ± 12	635 ± 14	2869 ± 14
R 9_35	382 ± 8.1	381 ± 8.1	3155 ± 16
R 11_45	493 ± 10	470 ± 10	2474 ± 12
R 1_55	439 ± 9.3	412 ± 8.7	16726 ± 84
R 4_05	514 ± 11	501 ± 11	2501 ± 13

^{x} Removed in later calculations due to large relative error.

A representative CA failure surface is shown in Fig. 6. This figure should show the fracture surface photographed after specimen failure and disassembly.

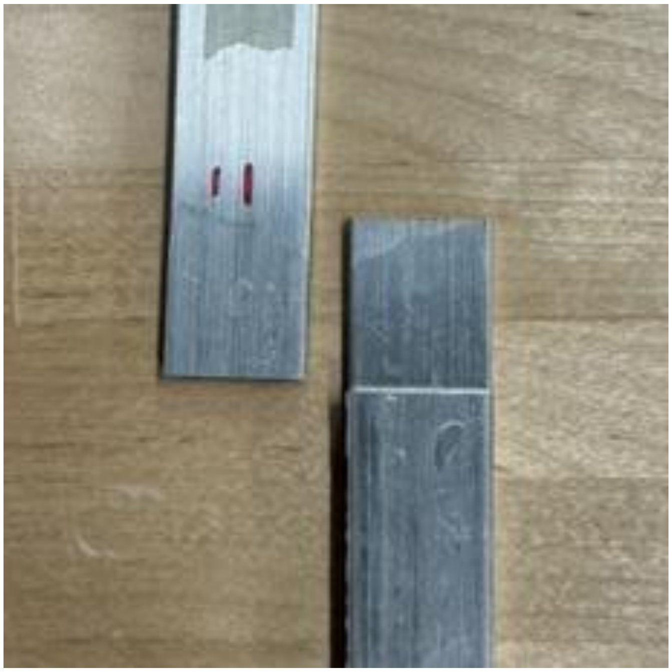

Fig 6. Failure surface of CA Adhesive from R1_55 Lab.

IV. Discussion

Using the measured maximum load and the measured failed bond area, the shear strengths shown in Tables V–VIII were computed from the raw measurements in Tables I–IV. The relative-error computation was utilized to find specimens that did not nearly meet the double-lap shear model's assumption that the two bonded sides carry equal loads for subsequent comparison. Large relative error specimens were eliminated from the subsequent statistical analysis, as the substrate-failure specimen as its highest load did not indicate adhesive shear failure. To keep the subsequent confidence-interval comparison linked to adhesive-joint shear behavior rather than highly asymmetric area measurements or non-adhesive failure modes, this filtering step was required.

TABLE V
Shear Strength and Relative Error for DP640 Adhesive

Lab Period	T_{Max} (MPa)	Relative Error %
T 9_35	4.8	0.34
T 11_45	4.7	1.5
T 1_55	6.0	5.4
T 4_05	2.3	0.14
W 10_40	2.0	7.3
W 12_50	3.5	7.3
W 3_00	0.95	0.19
R 9_35	3.0	3.0
R 11_45	2.9	0.38
R 1_55	3.6 ± 0.00011	0.30
R 4_05	2.6	2.3

TABLE VI
Shear Strength and Relative Error for AA H3000 Adhesive

Lab Period	T_{Max} (MPa)	Relative Error %
T 9_35	19	14
T 11_45	19	3.2
T 1_55	11	16
T 4_05	26	0.55
W 10_40	21	0.36
W 12_50	11	37
W 3_00	33	24
R 9_35	21	3.3
R 11_45	21	0.19
R 1_55	24 ± 0.00099	46
R 4_05	18	0.17

TABLE VII
Shear Strength and Relative Error for EA E60HP Adhesive

Lab Period	T_{Max} (MPa)	Relative Error %
T 9_35	4.4	7.9
T 11_45	10	0.53
T 1_55	2.6	6.5
T 4_05	6.5	14
W 10_40	8.6	6.0
W 12_50	6.3	5.5
W 3_00	8.5	3.1
R 9_35	12	2.1
R 11_45	7.5	0.64
R 1_55	6.3 ± 0.00020	2.0
R 4_05	3.6	5.1

TABLE VIII
Shear Strength and Relative Error for CA Adhesive

Lab Period	T_{Max} (MPa)	Relative Error %
T 9_35	2.1	0.74
T 11_45	3.3	35
T 1_55	3.6	19
T 4_05	3.0	0.22
W 10_40	2.4	1.1
W 12_50	2.8	14
W 3_00	2.4	5.4
R 9_35	4.1	0.15
R 11_45	2.6	2.4
R 1_55	20 ± 0.00069	3.2
R 4_05	2.5	1.3

That ranking is supported by the reported failure modes. Adhesive failure predominated in the lower-strength groups, suggesting that the adhesive-aluminum interface was typically the weakest point in those connections. AA H3000, on the other hand, had a wider variety of reported failure behavior, such as cohesive, mixed, and substrate-failure situations, indicating that the link at the interface was frequently strong enough for failure to move away from a purely interfacial separation. For the R 1_55 specimen, which was noted as a cohesive failure, this is particularly crucial. The higher measured shear strengths observed for AA H3000 overall are consistent with cohesive failure, which indicates that the adhesive itself became the limiting region rather than the adhesive-aluminum interface. Similarly, the substrate-failure specimen indicates that in at least one case the adherend failed before the adhesive joint, which prevented that test from serving as a clean adhesive-strength measurement. These observations strengthen the conclusion that AA H3000 developed the strongest bond condition in this experiment, while the other adhesives remained more interface limited.

TABLE IX
95% CONFIDENCE INTERVAL FOR MAXIMUM SHEAR STRESS

Adhesive	N	Average T_{Max} (MPa)	95% CI (MPa)
DP640	11	3.3	3.3 ± 0.96
AA H3000	7	20	20 ± 4.2
EA E60HP	11	6.4	6.4 ± 1.9
CA	10	4.5	4.5 ± 3.8

Additionally, the dispersion in the confidence intervals demonstrates that variation was not solely caused by measurement uncertainty. Because the load-cell accuracy was assumed to be 0.5% of the measured reading and the cleaned ImageJ dataset yielded a relative area uncertainty of only roughly 2.12% per measured area, the direct load uncertainty was minimal. Compared to the relative sizes of the statistical confidence intervals in Table IX, such direct uncertainties are less. For instance, for DP640, AA H3000, EA E60HP, and CA, the 95% CI half-width is roughly 29%, 18%, 30%, and 84% of the mean, respectively. Since the statistical spread is much larger than the direct uncertainty in force and area measurement, the dominant variation in the present data was more likely caused by specimen fabrication differences rather than by instrument limitations alone. Likely contributors include bond line thickness variation, slight alignment differences, unequal adhesive squeeze-out, local defects, and changes in how the joint failed from one specimen to another.

Another useful check on the values measured is the manufacturer comparison shown in Table X. With a measured mean of 20 MPa and a manufacturer reference of 20.3 MPa, AA H3000 demonstrated the best agreement with its reference value. On the other hand, EA E60HP measured 6.4 MPa against a reference value of 18 MPa, while DP640 measured just 3.3 MPa against a reference value of 14.9 MPa. The CA adhesive's measurement of 4.5 MPa was comparable to the comparison table's predicted reference of 4.4 MPa. While DP640 and EA E60HP performed worse than their stated reference strengths, the agreement for AA H3000 indicates that its performance in the current test was near to the anticipated aluminum lap-shear behavior. This discrepancy makes sense because the vendor data was collected under more regulated testing and preparation settings than those seen in the current lab. Since the current specimens were made from 6061-T6 aluminum, the measured strength represents the quality of the preparation, alignment, and curing that were accomplished during the lab procedure in addition to the adhesive qualities.

TABLE X
COMPARISON OF MANUFACTURER SHEAR STRENGTH OF ADHESIVES

Adhesive	Shear Strength	Units	Source
DP640	14.9	MPa	[4]
AA H3000	20.3	MPa	[5]
EA E60HP	18	MPa	[6]
CA	4.4*	MPa	[7]

It is important to recognize several limitations when interpreting these findings. First, the requirement to eliminate specimens for significant relative inaccuracy indicates that several joints did not sufficiently satisfy the equal-load assumption for the shear model to accurately depict the failure scenario. Second, the existence of substrate failure shows that adhesive failure was not necessarily forced by geometry prior to aluminum adherend failure. Third, the recorded fracture boundaries were sufficiently irregular that the area measurements introduced discernible errors to each shear-stress calculation, even if the ImageJ uncertainty was less than the class wide statistical spread. Lastly, because the manufacturer values were achieved on different aluminum conditions, with different surface preparations, and sometimes under different test standards than the current joints, the reference comparisons were simply approximations. For these reasons, rather than being universal values for every product, the results should be understood as the behavior of the adhesive systems under the settings of this lab.

Future phases of the experiment could benefit from several improvements. In addition to lowering the number of specimens discarded for significant relative error, a special assembly fixture would help in maintaining the center of the bonded sides. Squeeze-out and adhesive thickness variance would be lessened by a regulated bondline-thickness technique like spacers or shims. To apply sanding, cleaning, and cure handling more uniformly throughout all areas, surface preparation should also be more strictly standardized. The repeatability of the ImageJ readings would be enhanced by a fixed camera location and a calibration scale positioned in the same plane as the fracture surface. Finally, the joint geometry for the stronger adhesives could be adjusted so that adhesive failure occurs before substrate failure, since substrate failure prevents a clean measurement of adhesive shear strength. These changes would reduce scatter and make the comparison among adhesive types more dependable.

Overall, the most important finding of this experiment is that **AA H3000 was the strongest adhesive tested**, both statistically and observationally. Its high average shear strength, non-overlapping 95% confidence interval relative to the lower-strength groups, and occurrence of cohesive failure all indicate that it developed a stronger bonded joint than the other adhesives under the present lab conditions. DP640, EA E60HP, and CA all produced lower average strengths, and their overlapping intervals indicate that the differences among those lower-strength groups were less definitive. The combination of raw measurements filtered shear-stress values, confidence intervals, failure-mode observations, and manufacturer comparison therefore gives a consistent overall picture of adhesive performance in this lab.

V. Conclusion

This study evaluated the shear strength of four adhesives in 6061-T6 aluminum double-lap joints by combining measured failure loads with post-failure ImageJ area estimates. Among the adhesives tested, AA H3000 produced the highest average shear strength and the clearest statistical separation from the other groups, while DP640, EA E60HP, and CA formed a lower-strength group with greater overlap and larger relative influence from specimen-to-specimen variation. This pattern was confirmed by the failure data, which showed that cohesive behavior was present in the stronger AA H3000 joints, but adhesive failure predominated in the lower-strength groups, suggesting that interfacial bonding was frequently the limiting factor in those situations. The most significant significance of these findings is that specimen preparation, bonded-area symmetry, bond line consistency, and failure mode all significantly influence the strength that can be obtained in practice; adhesive selection alone does not dictate joint performance. For the measured shear strengths to reflect the intrinsic performance of each adhesive more accurately and to be more confident compared to reference data, future work should concentrate on improving assembly control through better fixtures, tighter surface-preparation standardization, controlled bond line thickness, and more repeatable post-failure imaging.

Appendix

A. ImageJ Failure-Area Uncertainty

The uncertainty in the ImageJ failure-area measurement was estimated from the repeated-measurement dataset obtained during the prelab activity. This dataset was used to quantify the scatter associated with repeatedly measuring the same failure area in ImageJ. The retained measurements were used to calculate the mean area, the sample standard deviation, the standard error of the mean, and the 95% confidence interval of the repeated-measurement dataset. Although the repeated-measurement confidence interval is useful for describing the spread of the dataset, the value adopted for later uncertainty propagation was the absolute area uncertainty based on two standard deviations of the final dataset. This follows the lab guidance that the expected uncertainty in the ImageJ area measurement may be estimated from the prelab data using $2s$. Thus, the repeated-measurement statistics were first calculated to describe the dataset, and then the final absolute ImageJ area uncertainty was taken as $u_A = 2s_A$, where u_A represents the uncertainty in one ImageJ area measurement. These are calculated using equations (6)-(11).

$$\bar{x} = \frac{1}{N} \cdot \sum_{i=1}^{N} x_i \tag{6}$$

$$S_x = \sqrt{\frac{1}{N-1} \cdot \sum_{i=1}^{N} (x_i - \bar{x})^2} \tag{7}$$

$$S_{\bar{x}} = \frac{S_x}{\sqrt{N}} \tag{8}$$

$$P_x = tS_x \tag{9}$$

$$CI = \bar{x} \pm P_{\bar{x}} \tag{10}$$

$$Bounds = \bar{x} \pm 2S_x \tag{11}$$

Using the final retained repeated-measurement dataset, the mean, standard deviation, and confidence interval were evaluated, and the adopted ImageJ uncertainty was then taken directly from Eq. (11). This value was used as the uncertainty for both measured failure-area terms in the later RSS propagation. The purpose of this subsection is therefore twofold: first, to document the statistical spread of the repeated ImageJ measurements, and second, to define the absolute uncertainty value later used in the propagation of total shear area and maximum shear stress.

B. Direct and Propagated Uncertainty Quantities

The uncertainty analysis for the present report was based on the direct uncertainty in the measured maximum load and the direct uncertainty in the two measured failure areas. The maximum load uncertainty was taken from the load-cell specification for the UTS, while the area uncertainty was taken from the repeated ImageJ analysis described in Appendix A. These direct uncertainties were then propagated through the total shear-area calculation and the maximum shear-stress calculation for the specimens in section R 1_55.

TABLE XI
Direct and Given Uncertainties For Adhesives

Name	Symbol	Uncertainty	Units	Source
Uncertainty of Load	μ_p	± 0.5	%	[2]
Uncertainty of Shear Area	μ_A	± 2.99	%	[RSS]
Uncertainty of Area A	μ_{AA}	± 2.12	%	[3]
Uncertainty of Area C	μ_{Ac}	± 2.12	%	[3]
Uncertainty of T_{Max}	μT_{Max}	± 0.005	MPa	[RSS]

Table XI summarizes the direct uncertainty in load and area and the propagated uncertainty in total shear area and maximum shear stress for the four specimens tested in section R 1_55. μ_p was taken from the load-cell specification, $u_{A_A} = u_{A_C} = u_A$ was taken from the ImageJ repeated-measurement analysis in Appendix A, $u_{A_{\text{shear}}}$ was calculated by RSS propagation, and $u_{\tau_{\max}}$ was calculated by RSS propagation of the shear-stress equation.

C. Root Sum Squared Propagation of Shear Area

The Root Sum Squared Method (RSS) of Uncertainty Propagation was applied to equation (2) to estimate the uncertainty in shear area μ_{As} from the uncertainty in the regions of each shear area μ_{AA} and μ_{Ac} as seen in equation 12.

$$\mu_{A_s} = \left[\left(\mu_{A_A}\frac{\partial A_s}{\partial A_A}\right)^2 + \left(\mu_{A_C}\frac{\partial A_s}{\partial A_C}\right)^2\right]^{\frac{1}{2}} \tag{12}$$

D. Root Sum Squared Propagation of Max Shear Stress

The measured maximum load and the measured failed shear area were used to compute the maximum shear stress. Since there is uncertainty in both the load and the area, the uncertainty in τmax must account for both terms' contributions. The UTS/load-cell specification provided the uncertainty in load, and the propagated value created in Appendix C provided the uncertainty in the failed area. RSS is applied to equation 1 to achieve this.

$$\mu_{\tau_{max}} = \left[\left(\mu_{A_s}\frac{\partial \tau_{max}}{\partial A_s}\right)^2 + \left(\mu_P\frac{\partial \tau_{max}}{\partial P}\right)^2\right]^{\frac{1}{2}} \tag{13}$$

E. Prelab ImageJ Measurements Used in Uncertainties

Table XII includes measured values from the prelab that were used to calculate the uncertainty of area using t-distributions.

TABLE XII
PRELAB IMAGEJ MEASUREMENTS

Areas Measured in mm^2							
5.2*	102.0*	102.0*	110.9*	117.0*	123.7*	150*	165*
165*	166*	170*	177*	178*	193*	252.0	275.0
300.0	300.0	300.0	300.0	315.0	354.0	369.0	375.0
384.3	400.0	400.0	400.0	400.0	400.0	410.0	410.0
413.1	420.0	420.0	420.0	420.0	420.0	423.0	426.1
430.0	430.0	432.0	434.0	434.0	434.0	437.5	438.5
439.2	440.0	440.7	442.6	445.3	446.1	447.0	447.3
447.5	450.0	450.0	450.0	450.0	452.0	453.4	456.0
456.0	459.2	460.0	460.0	460.1	461.0	461.4	462.5
464.8	465.1	466.0	466.0	467.5	468.6	469.0	470.7
471.0	471.1	475.0	475.5	476.0	477.0	477.0	479.7
480.0	480.0	480.0	480.5	480.6	481.0	482.0	483.7
484.6	485.0	485.1	486.0	487.2	487.8	487.8	488.5
491.0	491.0	491.7	492.0	492.2	492.7	493.0	493.0
493.0	494.0	494.2	494.4	494.5	495.9	497.0	497.0
498.3	498.8	499.8	499.8	499.8	499.8	499.9	500.0
500.0	500.0	500.1	502.0	503.8	504.7	506.0	506.2
506.5	506.5	507.3	508.0	509.1	514.9	515.3	516.0
516.0	516.0	516.2	517.0	518.0	518.6	519.5	520.6
520.7	523.4	524.1	525.1	525.6	527.0	528.1	529.3

TABLE XII CONT.
PRELAB IMAGEJ MEASUREMENTS

Areas Measured in mm^2							
530.0	530.6	532.1	532.4	533.7	537.4	539.0	544.6
548.0	548.2	549.2	549.9	550.0	550.0	550.0	550.0
551.0	551.5	552.5	552.6	557.3	558.3	559.0	559.3
560.0	560.4	560.8	563.0	566.3	576.5	578.2	579.0
579.7	580.0	580.8	587.2	587.3	590.3	591.0	591.9
593.0	595.0	597.1	599.5	600.0	600.0	610.0	614.0
617.0	620.0	620.4	625.1	631.7	634.0	636.2	640.0
650.0	650.0	655.3	670.0	670.0	670.0	674.3	687.6
689.3	693.2	698.6	700.0				

*indicates outliers that were removed from calculations.

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